From pluggable to co-packaged optics (CPO): an intro to AIDC interconnect

Oct 2025

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Why the focus on CPO?

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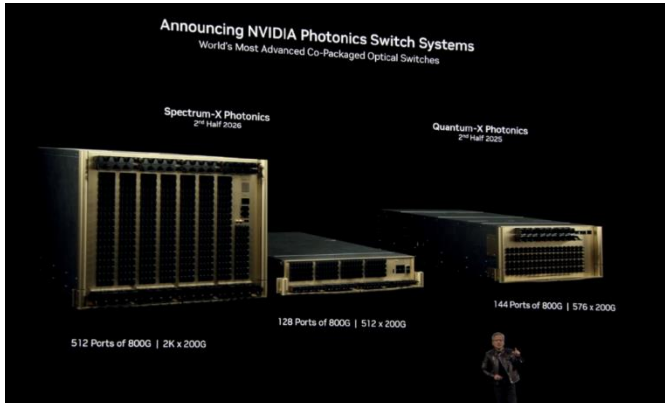
BROADCOM Tomahawk Switches

Accelerate CPO ecosystem adoption

- Tomahawk 4: 1st CPO switch chip announced in 2021
- Tomahawk 6: fastest CPO switch chip, announced sampling and testing in Oct. 2025

NVIDIA GTC 2025

Premiere advanced CPO switches



- Pivot from copper to optics
- Launch 2 CPO switches in Mar. 2025

deepseek 《Insights into DS-V3》

Highlight CPO's importance under limited bandwidth

6.3 Toward Intelligent Networks for AI

To meet the demands of latency-sensitive workloads, future interconnects must prioritize both low latency and intelligent networks:

- **Co-Packaged Optics:** Incorporating silicon photonics enables scalable higher bandwidth scalability and enhanced energy efficiency, both are critical for large-scale distributed systems.
- **Lossless Network**: Credit-Based Flow Control (CBFC) mechanisms ensures lossless data transmission, yet naively triggering flow control can induce severe head-of-line blocking. Therefore, it is imperative to deploy advanced, endpoint-driven congestion control (CC) algorithms that proactively regulate injection rates and avert pathological congestion scenarios.

- Bandwidth limitations restrict training, inference, and network scalability
- CPO could deliver higher bandwidth and enhance energy efficiency (published in May 2025)

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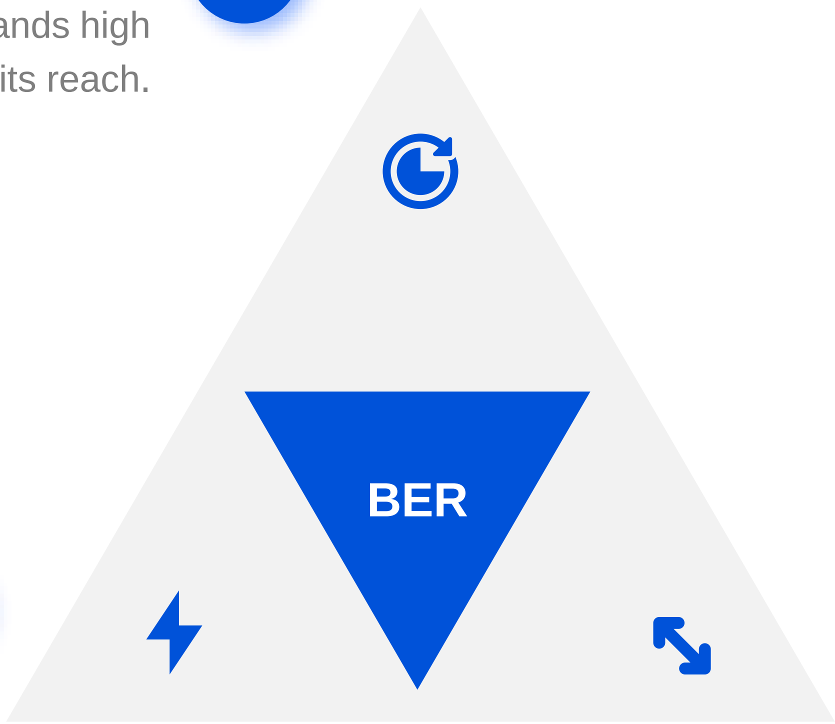
Communication trade-off triangle in IDCs: distance, power and speed

System goal: Given a **Bit Error Rate**[1] (BER, the core transmission reliability metric in IDC data communications) target, optimize the **tradeoff among speed, distance and power**

1 Distance

Long-reach transmission demands high power; low power limits reach.



Domestic markets prioritize speed over power (not a constraint).

2 Power

Long-distance suffers signal attenuation, requiring power compensation.

3 Speed

Low power + extended distance requires reduced data rates or bandwidth.

Note 1: BER standard for IDC scenarios is <1e-12, representing 1 bit error per trillion bits transmitted.

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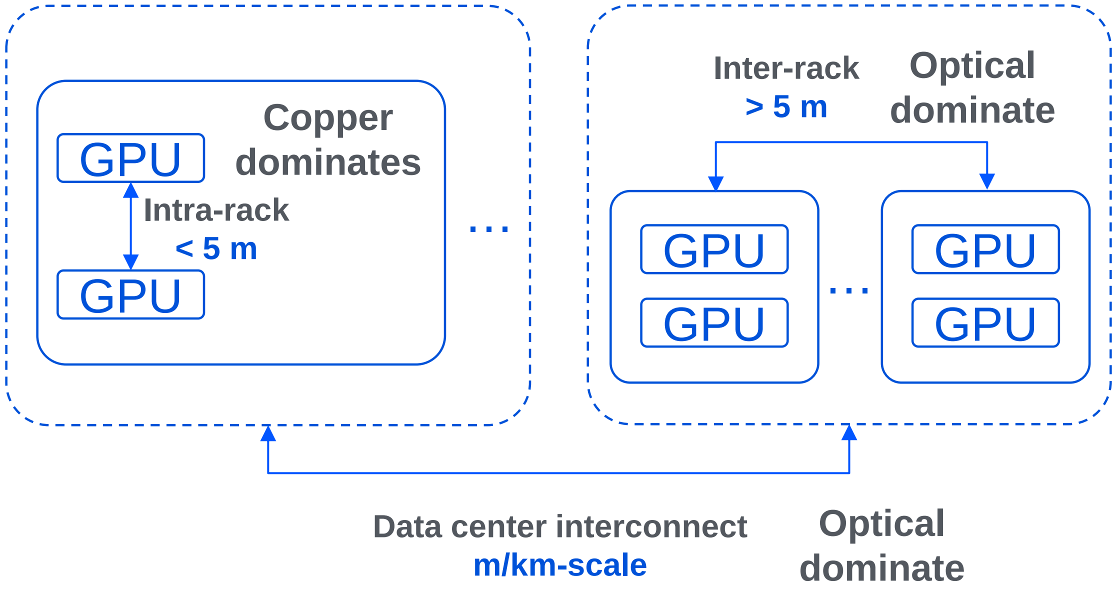
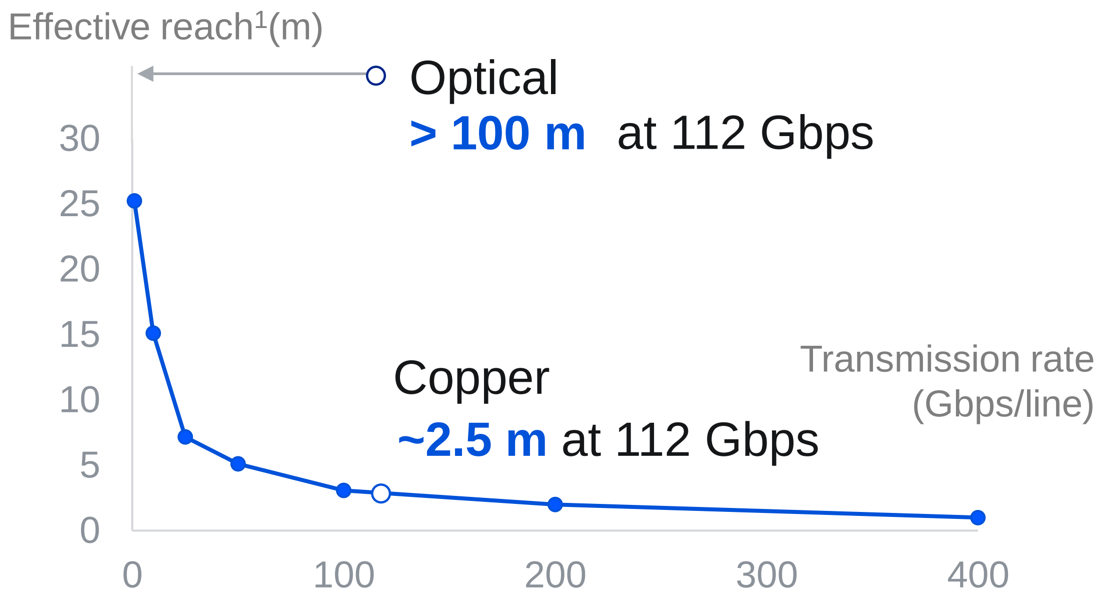
Optical interconnect overcomes the bottleneck of copper reach in IDC scaling

Expanding IDCs relies on optical interconnect



Optics significantly increases effective reach



Source: Marvell, Broadcom expert call, Microsoft Research.

Note 1: 112 Gbps is the current mainstream rate, while 224 Gbps is now emerging.

Note 2: The stable distance ensuring BER remains below a usable threshold.

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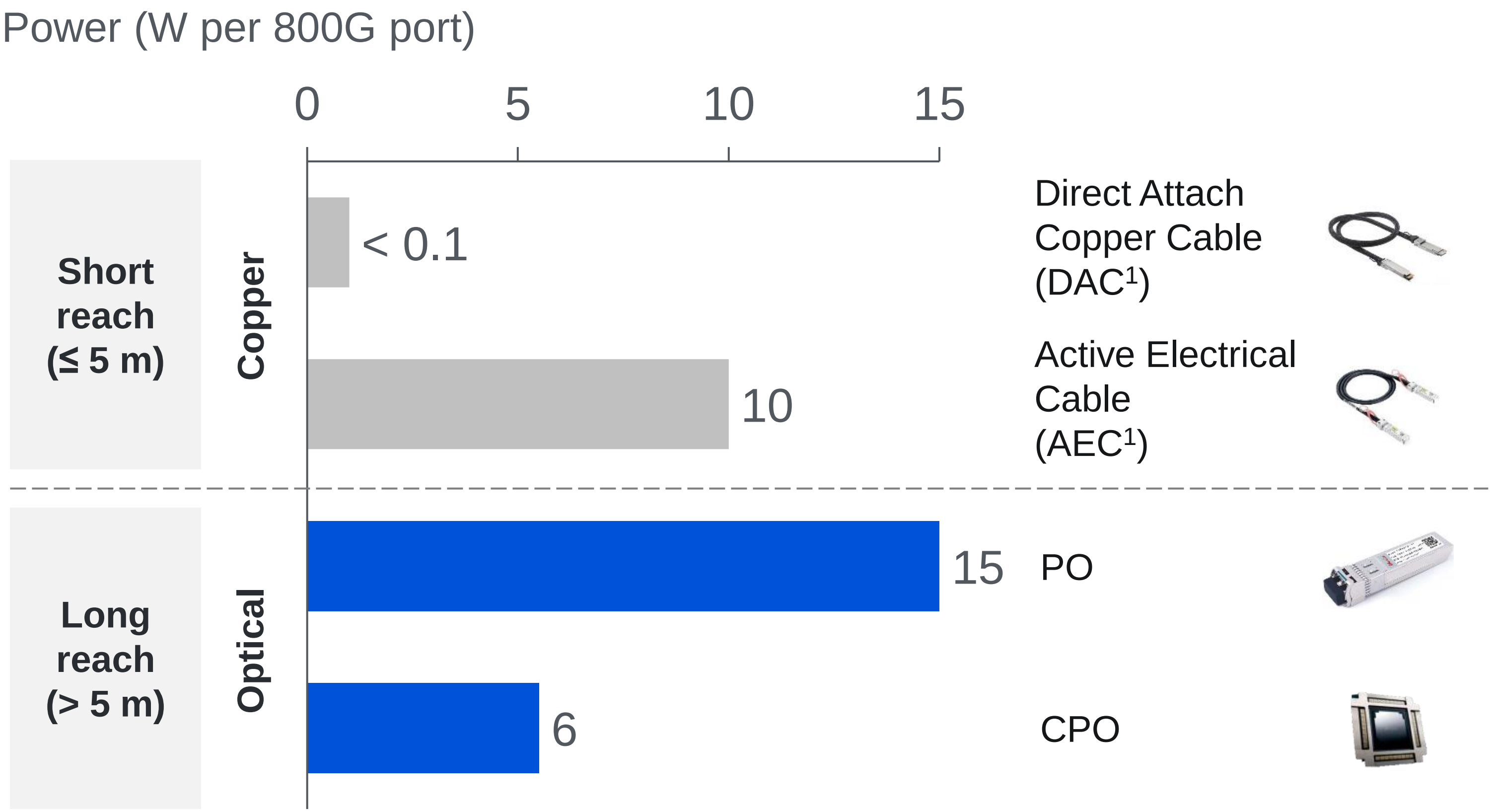
The power issue in the copper-to-optical migration is cut significantly by CPO

Power consumption: Copper, Pluggable, and CPO



- **Pluggable Optics (PO):** Entered mass production in 2014, accounting for **7.4%** of GB200 NVL72 card power[2].
- **Pluggable to CPO: ~40%** system-level power cut[3]; **5-mn RMB/y** saved for 10k-card cluster[2].

Source: LightCounting, Nvidia, Broadcom.

Note 1: The transmission reach of DAC is less than 2 meters. AEC integrates signal enhancement chips (re-timer) to extend the reach (5 meters) using more power.

Note 2: Consider GB200 NVL72 cluster with 9,216 B200 GPUs with 2-layer interconnect.

Note 3: At the 2025 Optica Executive Forum held in March, Broadcom demonstrated power savings of CPO in full 64-port system tests.

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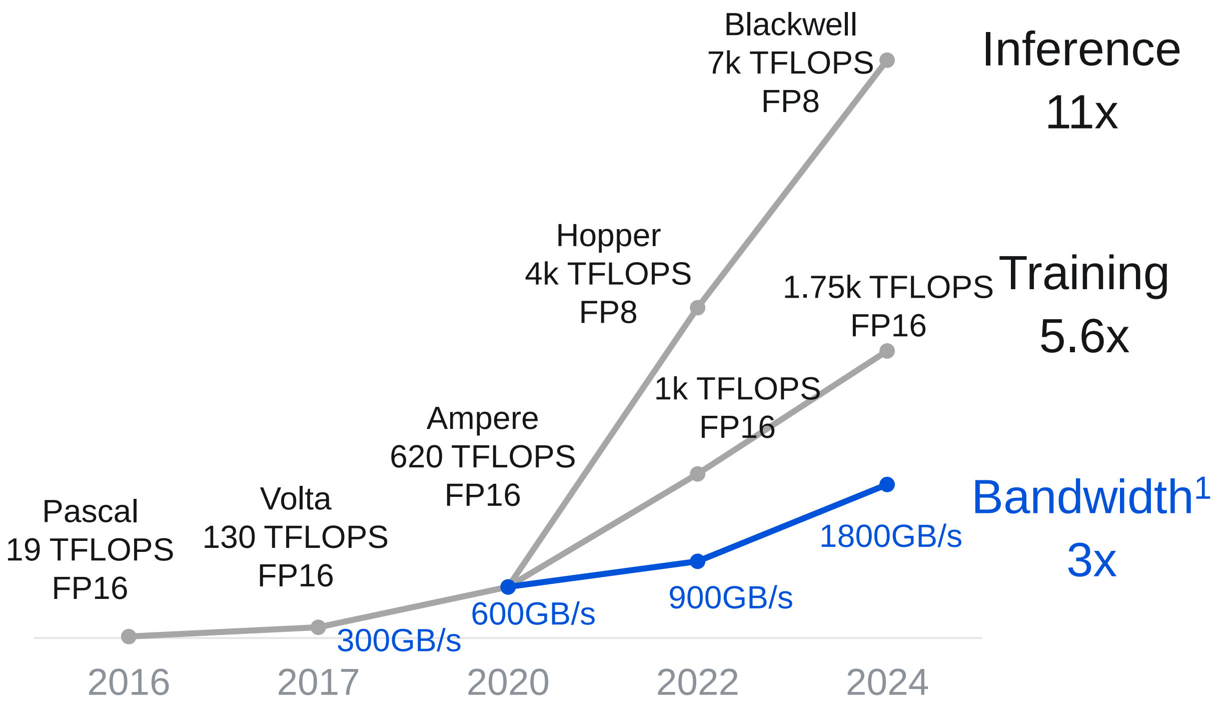
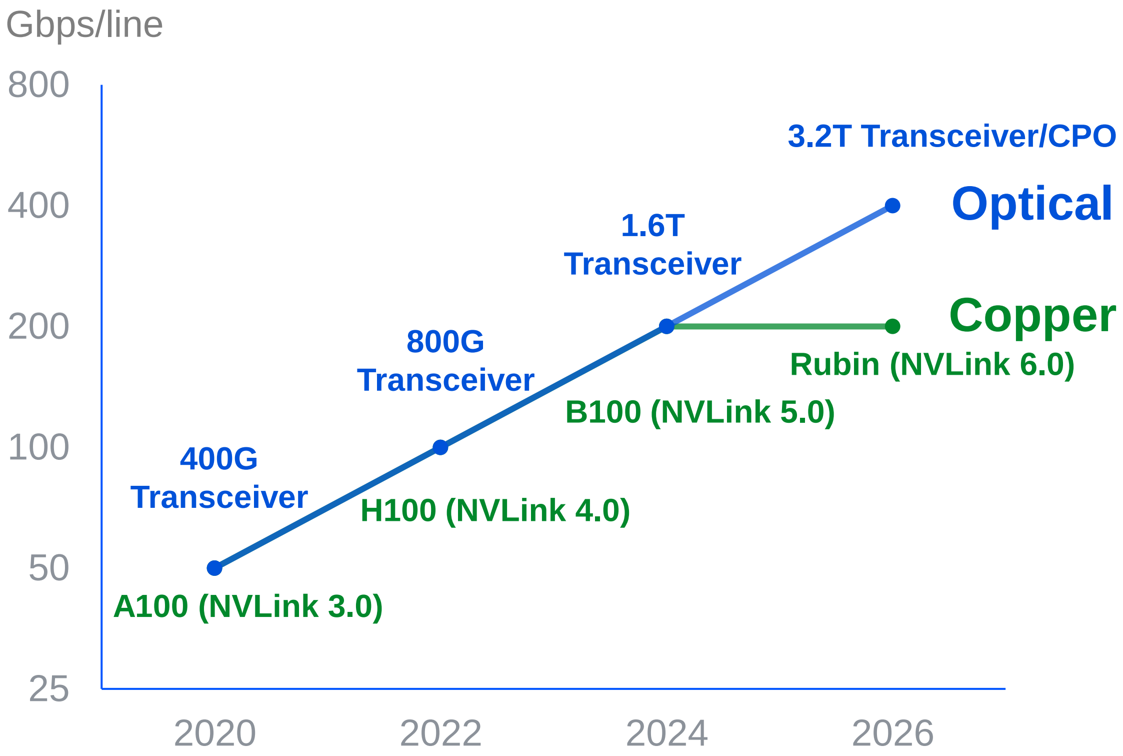
Optical interconnect is expected to surpass copper in transmission rate

Transmission rate growth lags compute



Optical at 448Gbps vs NVLink at 224Gbps



Source: Nvidia.

Note 1: NVLink bandwidth per card.

Note 2: At 224 Gbps, copper interconnects hit a wall, and unavoidable loss leads to unsustainable power and thermal costs.

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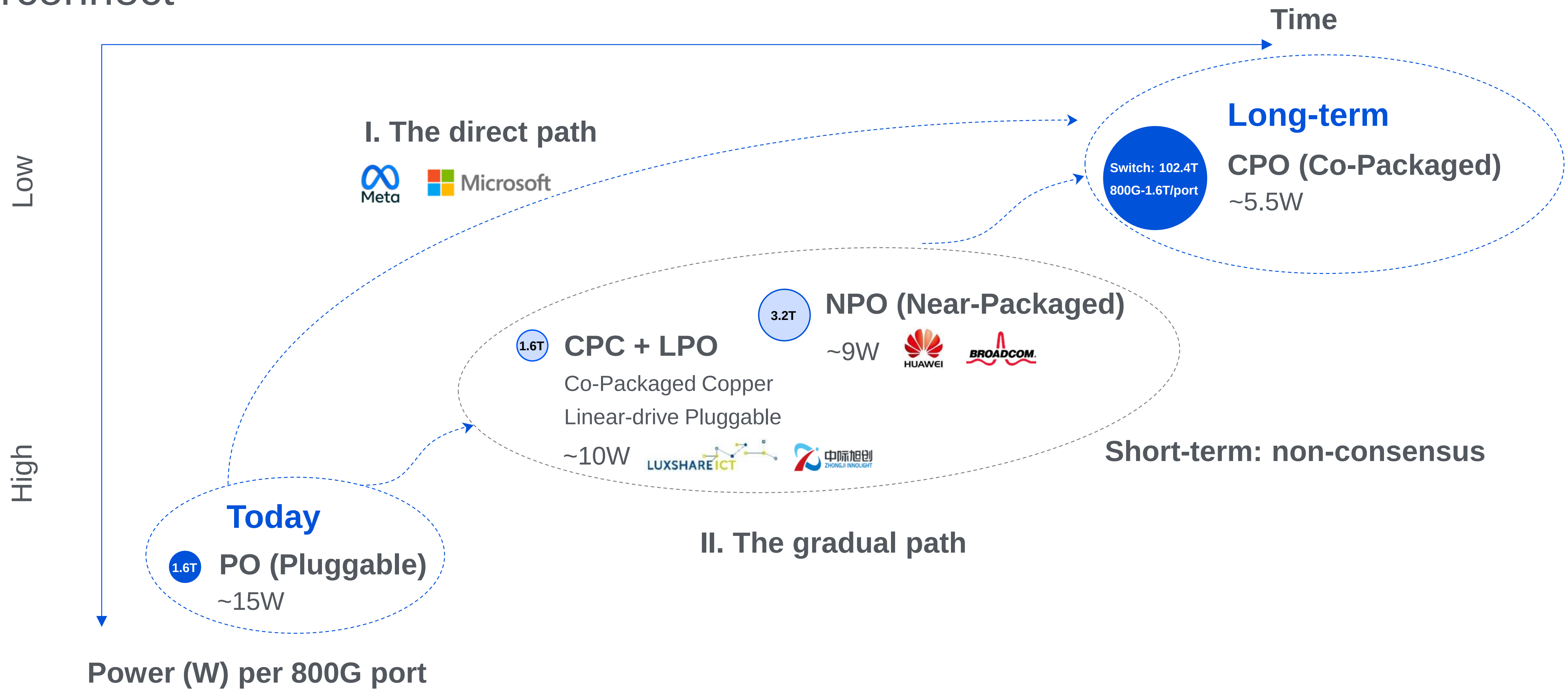
Copper-to-optical migration: players adopt different roadmaps within optical interconnect



Circle size denotes the latest data rate range.

Source: TSMC, Broadcom, Meta, Huawei

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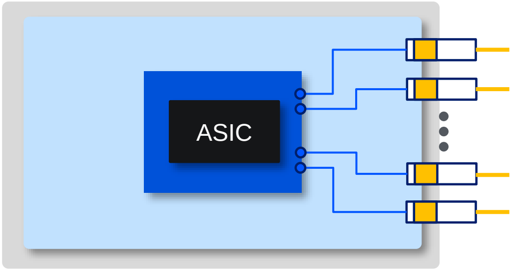
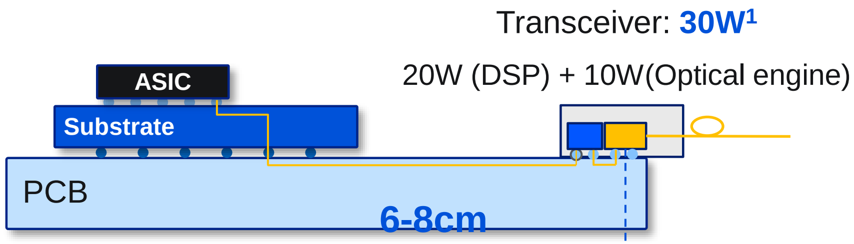
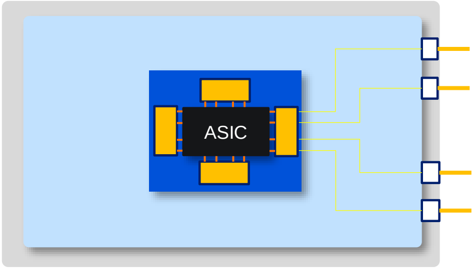
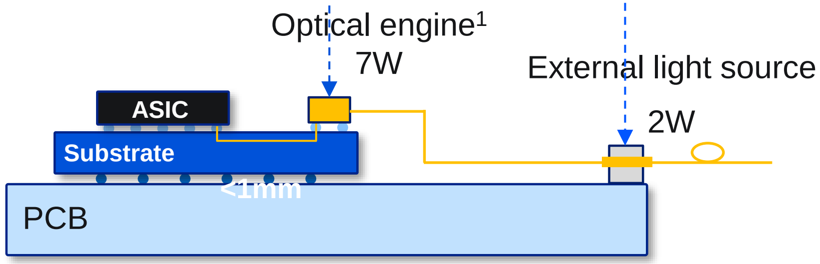
CPO overcomes PO’s constraints: signal loss and power consumption

CPO: By bypassing PCB traces, it reduces loss from 22 dB to **4 dB** and cuts power by **~60%** (port-level)

- **Hard constraint:** PCB-induced **loss/distortion becomes uncorrectable** beyond 224 Gbps/line
- **Soft constraint:** Long-reach transmission requires power compensation

	Signal loss	Front view	Side view
Pluggable	22dB		
CPO	4dB		

Co-packaging adds expense and maintenance challenges

Source: Nvidia, Broadcom.

Note 1: Power consumption in 1.6T port (from Nvidia).

If CPO becomes mainstream, global giants are expected to gain even greater influence in the new value chain

	PO (Pluggable Optics)	CPO (Co-Packaged Optics)
Total rate	**1.6T**bps	**6.4T**bps
Market size[1] 2025E	~$10bn	mn-scale
Key players[2]	**Domestic players dominate:** account for **70%** of the top 10 global vendors. 中际旭创 ZHONGJI INNOLIGHT, eoptolink, COHERENT 800G CR3: 80%	**Global giants bet on CPO:** viewed as the **most commercially viable path** for next-gen interconnect technologies. BROADCOM, NVIDIA
Dynamics	Mass commercial adoption in AIDC. 中际旭创 ZHONGJI INNOLIGHT • 2024: 800G shipments > 2mn units • Q4 25: 1.6T first to certify & ship	Targeting commercialization in the **25-26** timeframe. NVIDIA • Q4 25: Quantum-X CPO switch volume production • H1 26: Spectrum-X CPO switch volume production

Source: TSMC, Insight Partners, QYResearch, Morgan Stanley Research, LightCounting.
Note 1: PO market size from LightCounting; CPO data from Insight Partners.
Note 2: Showing only companies with the largest market share in each segment.

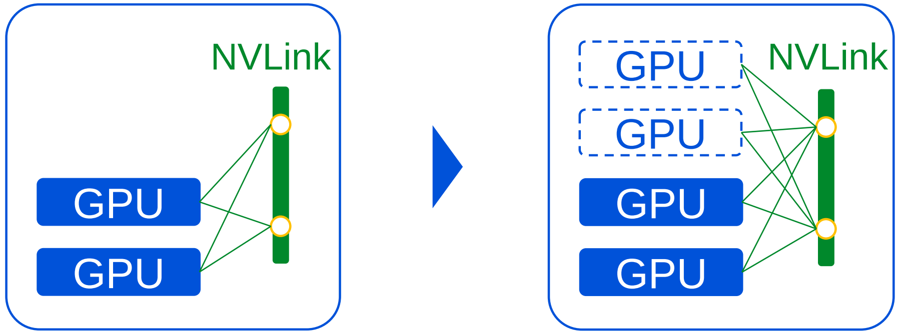
IDC interconnect can be categorized into scale-up and scale-out

Scale-up

- Single-node direct & full interconnect;
- Bandwidth: **TB/s**; Latency: **sub-μs**;
- Primarily **copper interconnect**.

Copper interconnect

Port (copper)



GB200 NVL72

B200 × **72 / rack**

Rubin NVL288

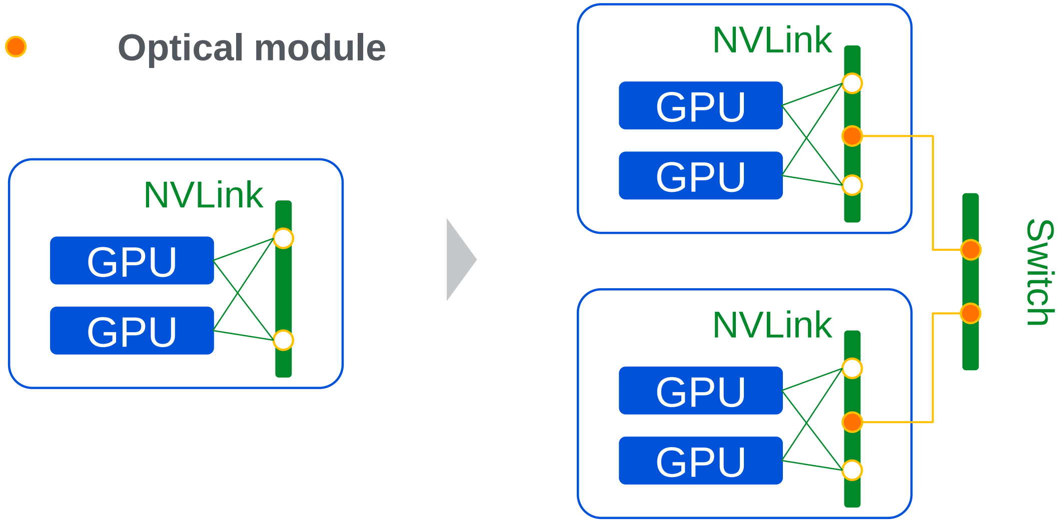
Rubin × **288 / rack**

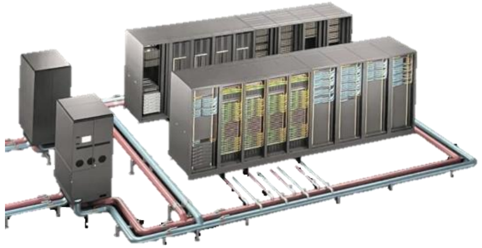
Scale-out

- Multi-node horizontal interconnect;
- Bandwidth: **GB/s**; Latency: **ms**;
- Primarily **optical interconnect**.

Optical interconnect

Optical module





GB200 SuperPod NVL 576

GB200 NVL72 × **8**

Source: Nvidia

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Optical interconnect adoption is prioritized in scale-out

	Overseas	Domestic
	No process constraints.	14nm & chip import ban
	Strong in electronics: BROADCOM, MARVELL — > 80% Switch/ DSP chips	Weak in electronics: Sitrus, Centec — < 5% Switch/ DSP chips[1]
Scale-out	**Initial adoption** • Transmission reach matters • Easier to commercialize	
Scale-up	**Acceleration** • Starting testing of optical in IDCs • Copper still dominates in the short-term	**Aggressive push** • Chip limitations drive longer-reach scale-up solutions • Weak in electronics, making optical essential for scale-up

Source: Changjiang Securities, Centec Communications IPO Prospectus, Sitrus Official Web Site.

Note 1: Sitrus and Centec are two leading Chinese companies specializing in high-speed DSP chips and switch chips, respectively.

Overseas players pursue scale-out and scale-up in parallel, with CPO initially focused on scale-out for 1-2 years

- **Volume production timeline: Scale-out** in **2026**; **Scale-up** in **2028** (per Broadcom)
- CPO adoption may **delay** due to gaps in supply chain **maturity** and **maintainability** compared to PO

NVIDIA

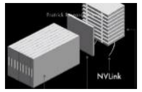
Full systems

BROADCOM

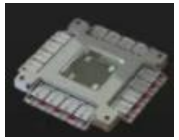
Switch chips

Scale-up

NVL 72 GB300
Rack-scale server
Q3 25: Volume production



NVL 576 Rubin
Rack-scale server
H1 26: Volume production



Tomahawk 6
Ethernet CPO chip
Oct. 25: Sampling and Testing

Scale-out

Quantum-X
InfiniBand CPO switch
Q4 25: Volume production

Spectrum-X
Ethernet CPO switch
H1 26: Volume production

Tomahawk 5
Ethernet CPO switch
H2 24: Sampling and Testing

xx Copper xx CPO

Source: Nvidia, Broadcom, Nomura research, BNP.

Note 1: Volume production date.

Estimating the CPO market: driven by port growth and data rate scaling

CPO scale[1] : ~$6 bn in 2028

Potential ports[2] ~ 57mn

800G+ transceiver shipments

GPU ports NVIDIA **~26 mn**

GPU shipments[2] **~5 mn**

Transceiver attach ratio[3] **~ 5 : 1**

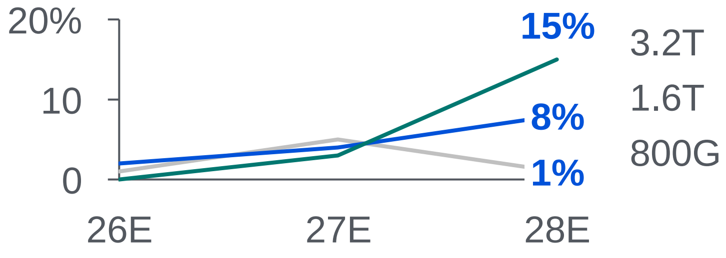
ASIC ports Google Cloud aws Meta **~31 mn**

ASIC shipment[4] **~6.3 mn**

Transceiver attach ratio[3] **~ 5 : 1**

Penetration rate[5] ~10%

800G+ transceivers => CPO



Unit Price[6] ~$1000

Per-port CPO value

Key assumptions

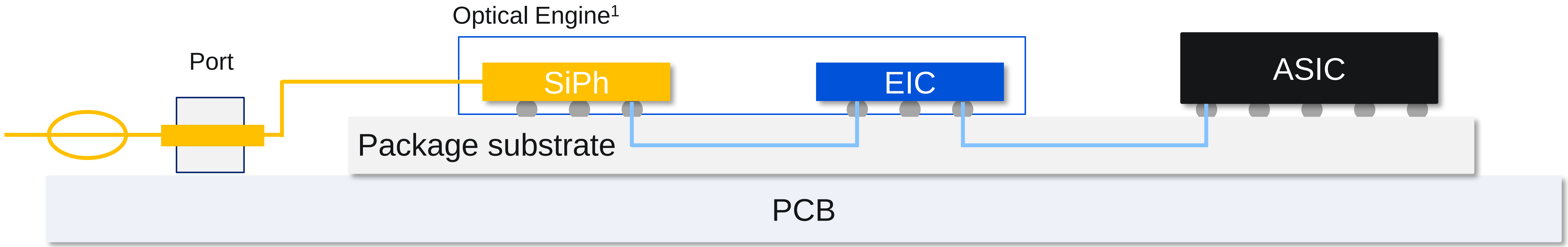
- Rubin (1.8 mn) and Rubin Ultra (3 mn) replace Hopper and Blackwell.
- 3 : 1 -> 5 : 1. Rubin requires twice the number of transceivers compared to GB300 (~3x 1.6T transceivers / card).
- ~55% of total datacenter AI chip volumes (Google TPU: > 4mn in 2028 with a shipment share of ~70%);
- 3 : 1 -> 5 : 1, driven by lower performance per card and increased cluster networking complexity.
- CPO starts penetration in late 2027: 1 year for mass production & half year for HPC adoption
- Nvidia Quantum CPO switch: $120-130k with 144 800G ports -> ~$900 / port in 2025;
- Rate 2x -> price 2x;
- Price descending accelerates in 2027 as CPO starts penetration.

Source: HSBC[2], Goldman Sachs[3, 4, 6], Citi[5].

Note 1: The model focuses primarily on the calculation logic, given that GPU/ASIC shipments and transceiver attach ratios are highly scenario-dependent.

CPO key modules: Silicon photonics (SiPh), electrical IC (EIC), switch ASIC, and integration & delivery

Input/Output	① Optical Engine: SiPh	② Optical Engine: EIC	③ Switch chip
Fiber	*Optical ↔ electrical signal conversion*	*Amplification + optimization*	*Control + signal processing*



④ Integration: advanced packaging + system integration

Optical signal

Electrical signal

Note 1: This is a simplified schematic emphasizing spatial relationships, not drawn to scale. The actual structure is more complex.

The supply chain is in early stage, with most players in initial strategic development

	① Optical Engine: SiPh — SiPh	① Optical Engine: SiPh — Laser	② Optical Engine: EIC	③ Switch chip	④ Integration — Switch	④ Integration — Assembly and test
Global leading players	Broadcom, NVIDIA, Marvell, intel, Cisco, Lumentum, Coherent	Broadcom, intel, Cisco, Lumentum, Coherent, MACOM	Marvell, Broadcom, Cisco, intel, NVIDIA, MACOM, Credo, Semtech	Broadcom, Marvell, NVIDIA, Cisco, intel, MediaTek	Cisco, Arista, NVIDIA, Huawei	tsmc, 日月光集團, Amkor Technology, JCET, Foxconn, fabrinet
Global key startups	Ayar Labs, celestial AI, Lightmatter, Ranovus	Lightmatter, sivers semiconductors	Litrinium, Retym	celestial AI, Salience Labs, Xsight	celestial AI, Micas	ficonTEC, Lightmatter
Domestic key players	中际旭创, eoptolink, Accelink, HGTECH, XPHOR 曦智科技, sicoya, SiLith, Tencent JDM on POs	Accelink, YJ, 飓芯科技 HURRICHIP	中晟微电子, Aluksen, Sitrus	Huawei, Clounix 云合智网, centec	Huawei, H3C, Ruijie 锐捷	ficonTEC, JCET, 日月光集團, ShnSin, TF 通富微电

Source: Yole

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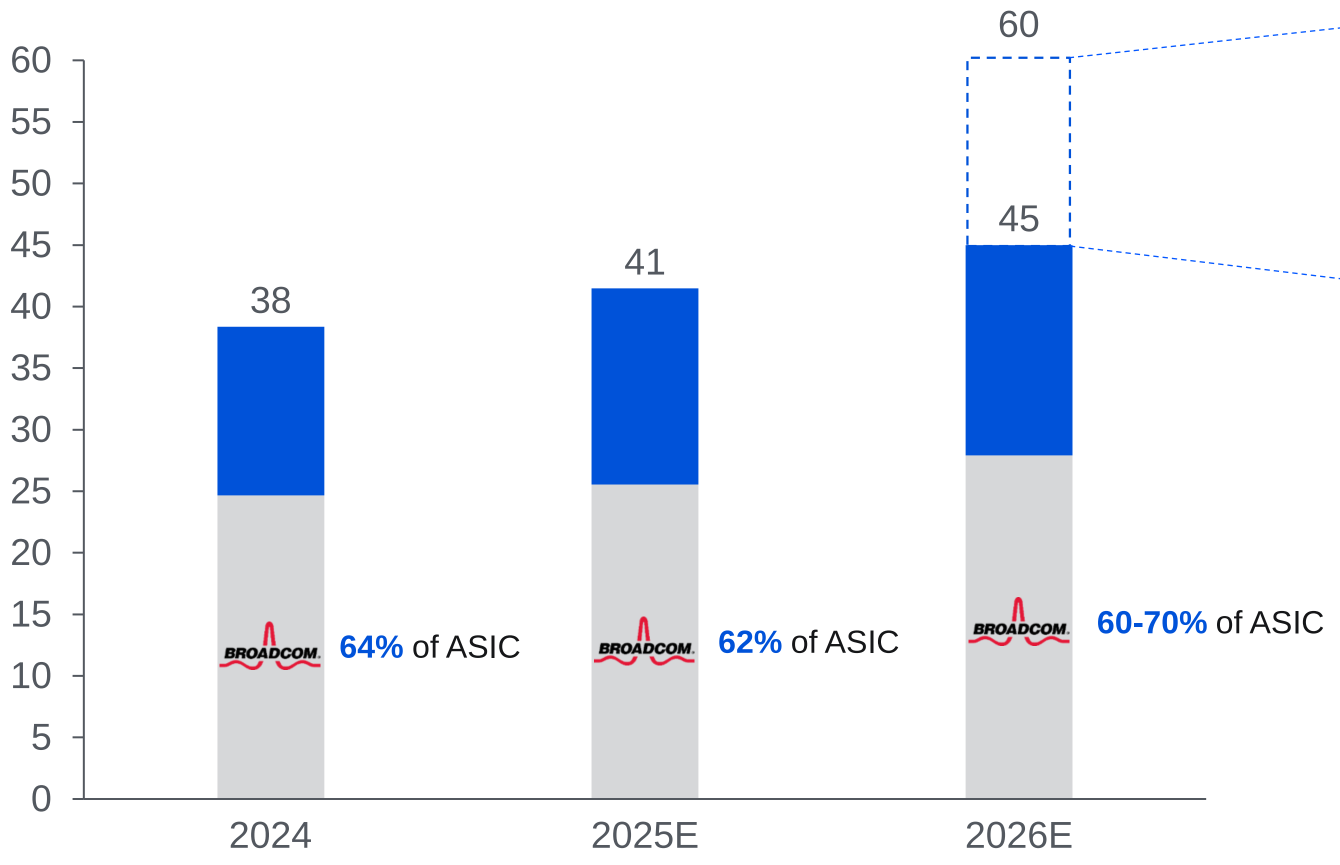
Broadcom's CPO bet: ASIC becomes the long-term trend, and Broadcom plays a central role

ASIC shipment unit share (%)



Optimistic scenario

- New orders from **OpenAI & xAI**;
- Bring forward the "ASIC > GPU" inflection point **from 2027–2028 to 2026.**

Pessimistic scenario

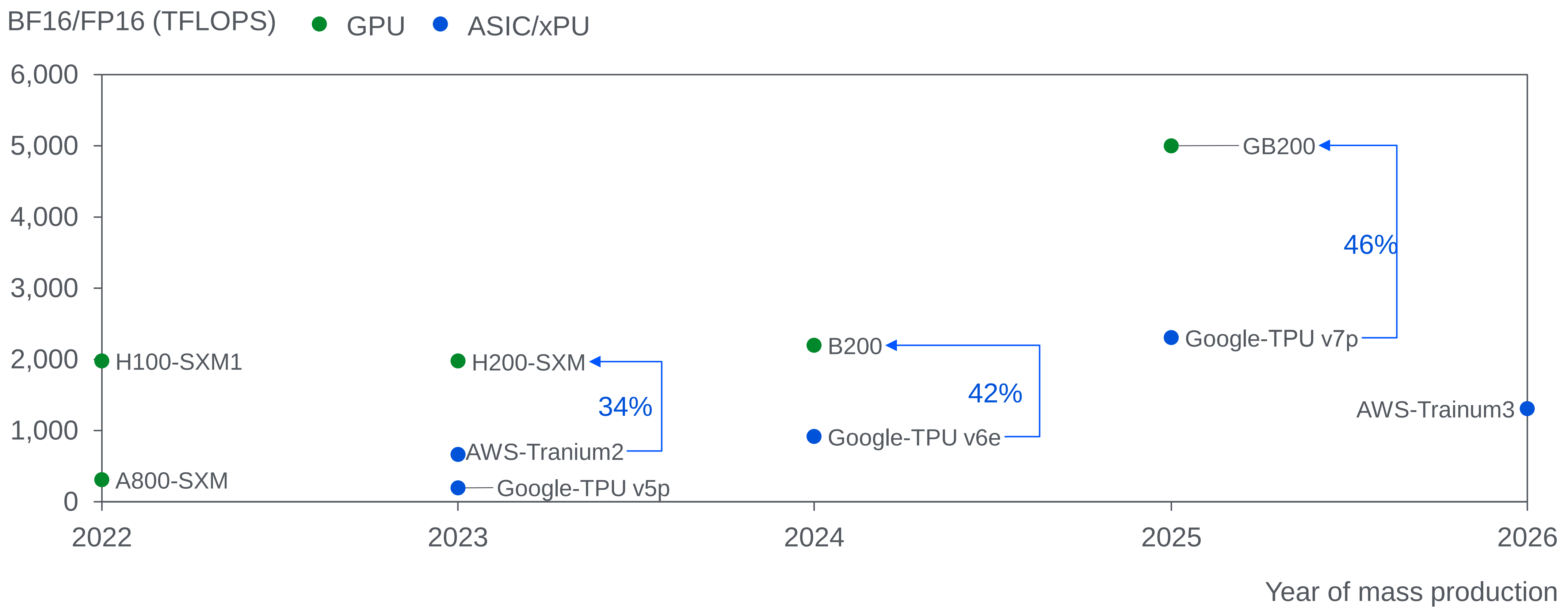
- ASIC: High costs, **limited TSMC supply**;
- **Nvidia continues to dominate** AI cloud compute.

Source: Nomura Securities, J.P.Morgan, Fubon Research, BNP.

Broadcom’s CPO bet: ASIC relies more on optical interconnect, and Broadcom attempts to lock in the data center ecosystem

ASIC: **Weaker per-card** performance, requiring optical interconnect (CPO) when scaling.

- Top-end ASICs deliver **~40%** of same-gen GPU throughput



Source: Fubon Research, Semiconductor Research, LightCounting.

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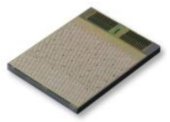
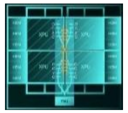
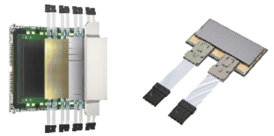
Startups with a long-term vision in optical: Ayar Labs, Celestial AI, and Lightmatter

	AyarLabs 2015 US	celestial AI 2020 US	LIGHTMATTER[1] 2017 US
Business & products	*Optical I/O (OIO) chiplets.*  • 512 Gbps OIO bandwidth • Ultra-low power: 4.9pJ/bit TeraPHY	*Optical full-stack solutions.*  • Card: PFLink，14.4 Tbps/die • Switch: PFSwitch Photonic Fabric	*CPO/OIO passage interposer.*  Passage Series • Proprietary silicon photonics chip • Package-level solution
Funding	1B Series D (Dec 24): $155mn	2.5B Series C1(Feb 25): $250mn	4.4B Series D (Oct 24): $400mn
Strategic investor	intel capital, NVIDIA, AMD, GlobalFoundries	AMD, SAMSUNG	GV
Advantages	• Technology integrated into PCIe 6.0 standard; • Ecosystem barrier.	• Systematic OIO with long R&D cycle; • Cross-layer solution.	• Compatible with Broadcom; • Abstracts packaging layer into hardware (controllable delivery).
Progress	POC -> Mass production • **Dec. 24:** Shipped **15k** chiplets in the past 18 months; • **26E:** First high-volume customers.	Certify -> Design-in / Early proto • **H2 25:** Sampling starts; • **27E:** Mass production.	Certify -> Design-in • **25 Summer:** Interposer shipment; • **26E:** L200 CPO chiplets shipment.

Note 1: Lightmatter (CEO: Nicholas Harris) and SuiJi Technology (a Chinese optical compute company invested by Tencent. CEO: Yichen Shen), spin-off from the same MIT team.

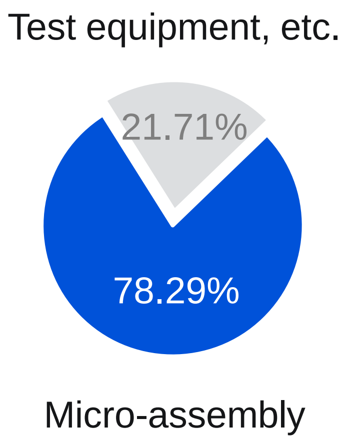
FiconTEC is a globally dominant player in SiPh packaging and testing

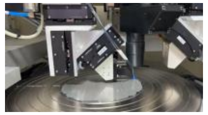
- **Supply chain position:** The only supplier in **fully-automated** SiPh/CPO assembly & testing;
- **Domestic substitution:** Fully **acquired by Chinese listed company** Robotechnik in 2025.

Business

- FiconTEC is the **key supplier of Nvidia, TSMC, and Broadcom** in SiPh/CPO assembly & testing.

Revenue breakdown (23A)





Test-to-repair All-in-one

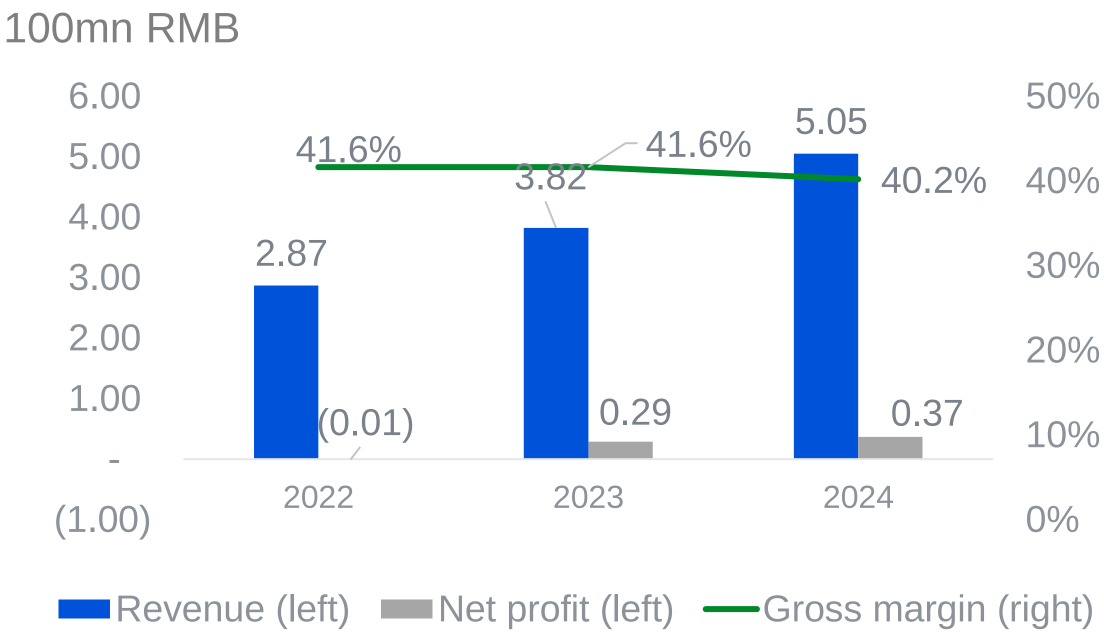
➡ **Vied for** by Nvidia and TSMC

AL-Series

➡ **Sole supplier** of Broadcom and Nvidia's CPO optical coupling equipment

Financials

- **~40% gross margin,** exceeding the industry average for assembly and testing (~30%).



Source: Robotechnik.

Summary

- **Trend:** **"Optics-in, Copper-out"** is the established path for high-speed AIDC interconnects

- **Scenario:** Optical interconnect gets **adopted first** in **scale-out**, with **accelerated penetration** in **scale-up**

- **Roadmap:** CPO is the **long-term bet**, while near-term solution is still not determined.

THANK YOU